Andy Baek

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Education

Boston University **Sep 2024 – Expected May 2028**
Bachelor of Science in Data Science *Boston, Massachusetts*
GPA: 3.8/4.0 (Dean's List)

Experience

BU Psychology Lab **Sept 2026 – Present**
Software Engineer | Next.js, React Native, FastAPI, Supabase *Boston, MA*

- Building a full-stack experimental platform with a 3-person team for a media psychology lab studying short-form video engagement, pairing a Next.js researcher web app with a React Native Android tablet app that mimics Reels/Shorts-style interfaces.
- Co-developing the React Native tablet app with a teammate and helping integrate both the web and tablet clients with the FastAPI + Supabase (PostgreSQL) backend.
- Implemented millisecond-precision behavioral event capture in the tablet app (watch time, swipes, button presses, survey first-touch) with video pre-loading to keep timing data valid for human-subject research.
- Set up the testing and CI pipeline from scratch with the team: Jest + React Testing Library component tests, Playwright end-to-end tests, pytest for the FastAPI backend, and GitHub Actions workflows that run linting, test coverage, and builds.

Boston University **Sept 2026 – Present**
Course Assistant, DS120 (Foundations of Data Science) *Boston, MA*

- Hold weekly office hours supporting students through proof-based coursework spanning mathematical logic, linear algebra, multivariable calculus, and probability theory.
- Grade weekly problem sets and exams for a 180+ student course, applying rubrics consistently across topics including linear systems, gradient descent, and hypothesis testing.

Hyundai Motor Manufacturing Alabama, LLC **June 2024 – Aug 2024**
Inventory Auditor *Montgomery, AL*

- Performed plant-wide inventory verification on a 1,500 vehicles/day assembly line, serving as the final check after section-level inspections and enabling same-shift restocking.
- Coordinated with materials and production teams to resolve inventory discrepancies in real time, preventing line stoppages during high-volume shifts.

Projects

Tennis Player & Ball Tracker | *Python, YOLOv8, OpenCV* **Sept 2025**

- Built an end-to-end video processing pipeline that ingests full-length match footage, segments it into live-play rallies, and outputs a condensed highlight video with tracking stats overlaid, reducing footage by 70%.
- Fine-tuned YOLOv8 for ball detection, raising detection from near-zero to **80–90%** of frames, and filled remaining gaps with trajectory interpolation for continuous tracks.
- Mapped player and ball positions onto a top-down 2D mini-court using court keypoint detection and a **homography**, powering automated shot and rally analysis.

Neural Network from Scratch in Rust | *Rust* **April 2025**

- Implemented a fully connected neural network in Rust without ML frameworks, writing the forward pass (ReLU hidden layer, softmax output) and cross-entropy loss directly on `ndarray` matrices.
- Derived and hand-coded backpropagation for all weights and biases, using the combined softmax + cross-entropy gradient $(\hat{y} - y)$ and batch-averaged mini-batch gradient descent updates.
- Designed the network around column-major batched inputs with broadcasted bias addition, so the same forward/backward code handles single samples and mini-batches.

Flappy Bird RL – Tabular Q-Learning vs. DQN with Behavioral Cloning | *Python, Gymnasium, PyTorch* **Mar 2026**

- Built a custom Flappy Bird environment (Pygame + Gymnasium) and implemented tabular Q-learning and DQN agents in PyTorch to compare classical vs. deep RL on the same task.
- Diagnosed a sparse-reward failure where agents rarely cleared the first pipe; added a gap-alignment reward-shaping term that took the final agent from failing at the first obstacle to surviving indefinitely (1,000+ pipes without failure).

Relevant Coursework

CS: Data Structures, Algorithms, Concepts of Programming Languages, Database Design, Cryptography
AI/ML & Math: Neural Networks, Reinforcement Learning, Linear Algebra, Probability & Statistics

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, Rust, OCaml, HTML/CSS, Shell/Bash
Frameworks & Libraries: React, Next.js, React Native (Expo), Node.js, FastAPI, PyTorch, OpenCV, NumPy, pandas
Testing & CI/CD: Jest, React Testing Library, Playwright, pytest, GitHub Actions, ESLint, Prettier
Tools & Platforms: PostgreSQL (Supabase), MongoDB, Docker, REST APIs, Postman, Git/GitHub, Linux/Unix, Vercel, LaTeX
Practices: Agile/Scrum, Git workflows (pull requests, code review), CI/CD